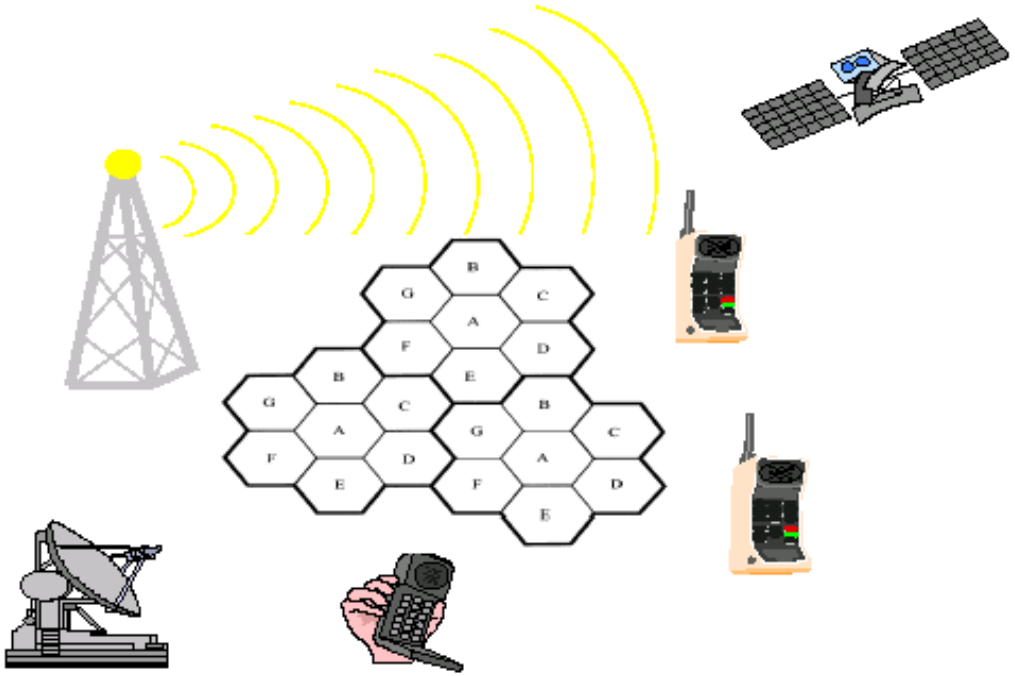
Chapter 1
INTRODUCTION TO WIRELESS COMMUNICATION SYSTEMS



Wireless Communication

Indu Bhardwaj
Bharati Vidyapeeth College of Engineering.
New Delhi, India
indubhardwaj2011@gmail.com

UNIT – I

Introduction To Wireless Communication Systems:

Evolution of mobile radio communications; examples of wireless comm. systems; paging systems; Cordless telephone systems; overview of generations of cellular systems, comparison of various wireless systems.

Introduction to Personal Communication Services (PCS):

PCS architecture, Mobility management, Networks signaling. basic cellular system, multiple access techniques: FDMA, TDMA, CDMA.

Introduction to Wireless Channels and Diversity:

Fast Fading Wireless Channel Modeling, Rayleigh/Ricean Fading Channels, BER Performance in Fading Channels, Introduction to Diversity modeling for Wireless Communications

Suggested Reading

Text Books:

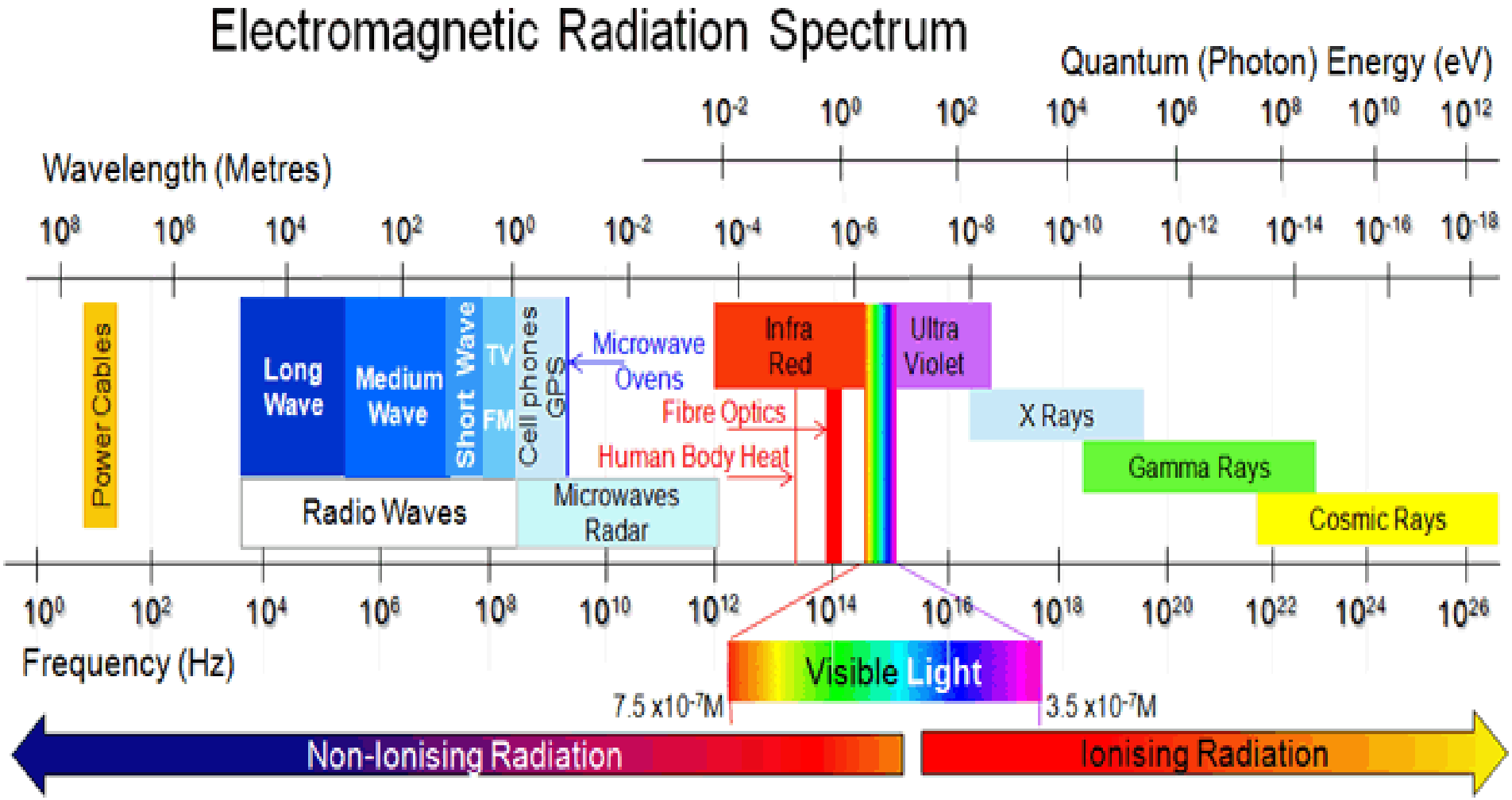
- Raj Pandya, "Mobile & Personnel communication Systems and Services", Prentice Hall India, 2001.
- Theodore S. Rappaport, "Wireless Communication- Principles and practices," 2nd Ed., Pearson Education Pvt. Ltd, 5th Edition, 2008.

Reference Books:

- T.L.Singhal "Wireless Communication", Tata McGraw Hill Publication.
- Jochen Schiller, "Mobile communications," Pearson Education Pvt. Ltd., 2002.
- Yi –Bing Lin & Imrich Chlamatac, "Wireless and Mobile Networks Architecture," John Wiley & Sons, 2001.
- Lee, W.C.Y., "Mobile Cellular Telecommunication", 2nd Edition, McGraw Hill,1998.
- Smith & Collins, "3G Wireless Networks," TMH, 2007
- Schiller, Jochen, "Mobile Communications", 2nd Edition, Addison Wesley

What is wireless communication?

- wireless communication is basically **transmitting and receiving voice and data** using electromagnetic waves in open space. They are basically free from wires.
- Electromagnetic waves
  - Travel at speed of light ($c = 3x10^8$ m/s)
  - Has a frequency (f) and wavelength (λ)
    - » $c = f \times \lambda$
  - Higher frequency means higher energy photons
  - The higher the energy photon the more penetrating is the radiation

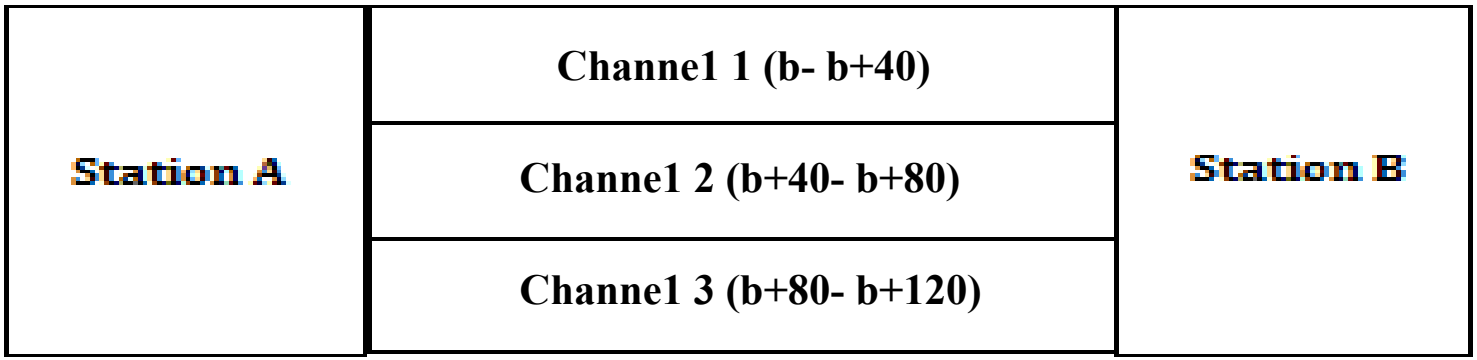

Electromagnetic Radiation Spectrum
Quantum (Photon) Energy (eV)
10-2 10^0 10^2 10^4 10^6 10^8 10^10 10^12
Wavelength (Metres)
10^8 10^6 10^4 10^2 10^0 10-2 10-4 10-6 10-8 10-10 10-12 10-14 10-16 10-18
Power Cables
Long Wave
Medium Wave
Short Wave
TV
FM
Cell phones
GPS
Microwave Ovens
Infra Red
Ultra Violet
Fibre Optics
X Rays
Human Body Heat
Gamma Rays
Radio Waves
Microwaves
Radar
Cosmic Rays
10^0 10^2 10^4 10^6 10^8 10^10 10^12 10^14 10^16 10^18 10^20 10^22 10^24 10^26
Frequency (Hz)
Visible Light
7.5 x10-7M
3.5 x10-7M
Non-Ionising Radiation
Ionising Radiation

Frequency carrier/channel

- The information from **sender to receiver** is carrier over a well defined frequency band.
  - ➢This is called a channel
- Each channel has a **fixed frequency bandwidth** (in KHz) and Capacity (bit-rate)
- Different frequency bands (channels) can be used to transmit information in **parallel** and independently.

Example

- Assume a spectrum of 120KHZ is allocated over a base frequency for communication between stations A and B
- Each channel occupies 40Khz

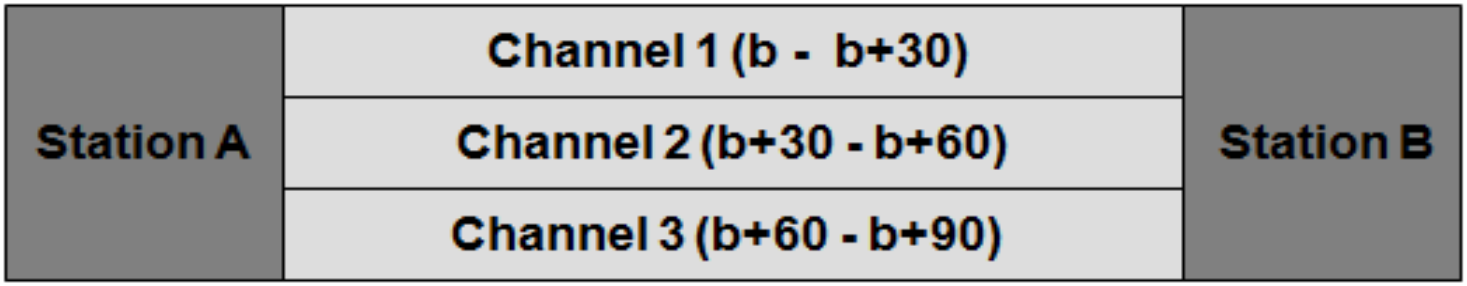


Explanation

- Suppose, we have an allocation of about 120 KHz of bandwidth and we require to communicate from station A to station B, a very simple way is to divide the entire bandwidth into 3 sub bands. Each one is known as a channel. Channel 1, channel 2 and channel 3 share about 40 KHz of bandwidth. In this idealized situation, station A can communicate through this three channels without the fear of interference.
- But practically there is problem of interference. To overcome this problem some space between the frequency allocations which are known as the guard bands are allocated . For example, channel 1 and channel 2 are not just adjacent to each other but they have a small guard band, let's suppose a 200 Hz of guard band in a40 KHz band.

Another example

- Assume a spectrum of 90KHz is allocated over a base frequency b for communication between stations A and B
- Assume each channel occupies 30KHz.
- There are 3 channels
- Each channel is simplex (Transmission occurs in one way)
- For full duplex communication:
  - Use two different channels (front and reverse channels)
  - Use time division in a channel



Typical Frequencies of some technologies

- FM radio 88MHZ
- TV broadcast 200 MHZ
- GSM phones 900MHZ
- GPS 1.2 GHZ
- PCS phones 1.8 GHZ
- Bluetooth 2.4GHZ
- Wifi 2.4GHZ

Why Wireless communication?

- **Freedom from wires**
  - no cost of installing wires or rewiring,
  - no stolen wires, no bunch of wires running here and there.
  - "Auto magical" instantaneous connection without physical connection setup e.g. Bluetooth Wifi
- **Global coverage**
  - Communication can reach where wiring is infeasible or costly e.g. rural areas, old buildings, vehicles, battle fields, outer space (through communication satellites)

Why Wireless communication?

- **Stay connected**
  - Roaming allows flexibility to stay connected anywhere and anytime
  - Rapidly growing market atleast to the public needs for mobility and uninterrupted access
- **Flexibility**
  - Services reach you wherever you go **(mobility).** E.g. you don't have to go to your lab to check your mail.
  - Connect to **multiple devices** simultaneously (no physical connections required)

Why Wireless communication?

- Increasing dependence on telecommunication services **for business and personal reasons**
- Consumers and businesses are willing to pay for it.

- **Basic Mantra: Stay Connected – anywhere, anytime.**

Wireless vs. mobile

- *NOTE:* wireless does not necessarily mean mobile
- Wireless system may be

  - Fixed (e.g. Metropolitian Area Network)

  -Mobile (mobile phone)

Wireless system classification

- Simplex Systems
  - Communication systems which provide only one-way communication
  - Pagers
- Half Duplex Systems
  - Communication Systems which allow two-way communication by using the same radio channel for both transmission and reception. At any given time, the user can either transmit or receive information.
  - Push-to-talk and release-to-listen systems
- Full Duplex Systems
  - Communication systems which allow simultaneous two-way communication. Transmission and reception is typically on two different channels (FDD).

Development of Mobile radio communication

- In 1897, Guglielmo **Marconi** first demonstrated radio's ability to provide continuous contact with ships sailing the English channel.
- During the past 10 years, fueled by
  - Digital and RF circuit fabrication improvements
  - New VLSI technologies
  - Other miniaturization technologies

    (e.g., passive components)

  The mobile communications industry has grown by orders of magnitude.
- The trends will continue at an even greater pace during the next decade.

Growth of mobile telephony

- compared with other inventions of 20th century

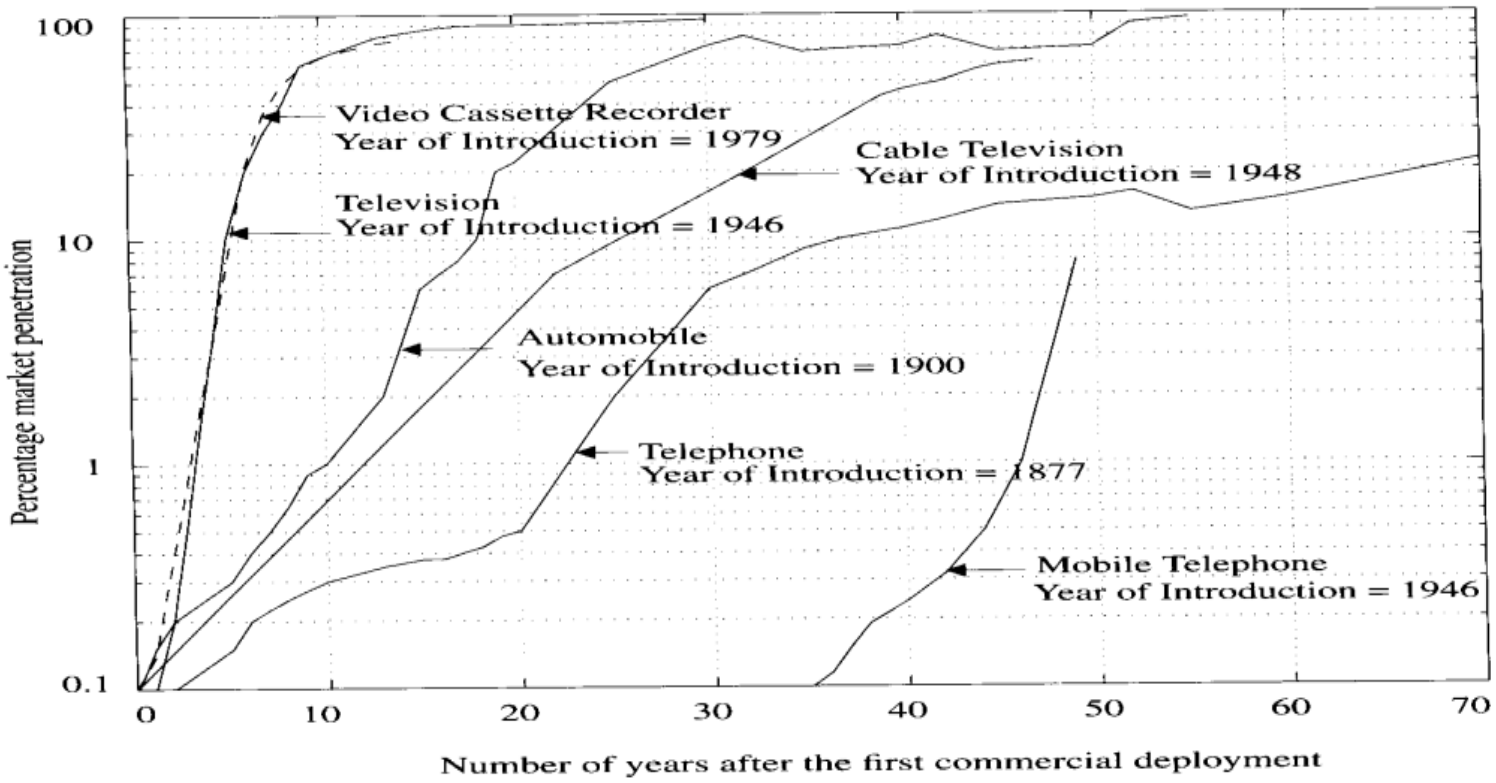


Figure 1.1
Figure illustrating the growth of mobile telephony as compared to other popular inventions of this century.

Evolution of Mobile Radio Communications

- In 1934, **AM mobile communication systems was adopted for municipal police radio systems** for public safety in US.
  - vehicle ignition noise was a major problem.

- In 1935 **Frequency modulation was demonstrated** by E. H. Armstrong.

- In 1946, FM mobile communications for **the first public mobile telephone service**
  - Each system used a single, high-powered transmitter and large tower to cover distances of over 50 km.
  - Used 120 kHz of RF bandwidth in a **half-duplex mode**. (push-to-talk release-to-listen systems.)

Evolution of Mobile Radio Communications

- In 1950, the channel **bandwidth was cut in half to 60kHZ** due to improved technology

- By the mid 1960s, the channel **bandwidth again was cut to 30 kHZ**. Thus, from WWII to the mid 1960s, the spectrum efficiency was improved only a factor of 4 due to the technology advancements.

- Also in 1950s and 1960s, **Automatic channel truncking** was introduced in IMTS(Improved Mobile Telephone Service.)
  - offering **full duplex**, auto-dial, auto-trunking
  - became saturated quickly
  - By 1976, has only twelve channels and could only serve 543 customers in New York City of 10 millions populations.

Evolution of Mobile Radio Communications

- In 1960s, **Cellular radio telephone** Developed by Bell Lab and others
  - The basic idea is to reuse the channel frequency at a sufficient distance to increase the spectrum efficiency.
  - But the technology was not available to implement until the late 1970s. (mainly the microprocessor and DSP technologies.)

- In 1983, **AMPS (Advanced Mobile Phone System, IS-41)** deployed by Ameritech in Chicago.
  - 40 MHz spectrum in 800 MHz band
  - 666 channels (+ 166 channels), per Fig 1.2.
  - Each duplex channel occupies > 60 kHz (30+30) FDMA to maximize capacity.
  - Two cellular providers in each market

Evolution of Mobile Radio Communications

- In late 1991, **U.S. Digital Cellular (USDC, IS-54)** was introduced.
  - to replace AMPS analog channels
  - 3 times of capacity due to the use of digital modulation (DQPSK), speech coding, and TDMA technologies.
  - could further increase up to 6 times of capacity given the advancements of DSP and speech coding technologies.

- In mid 1990s, Code Division Multiple Access **(CDMA, IS-95) was introduced by Qualcomm.**
  - based on spread spectrum technology.
  - supports 6-20 times of users in 1.25 MHz shared by all the channels.
  - each associated with a unique code sequence.
  - operate at much smaller SNR.(FdB)

Evolution of Mobile Radio Communications

- **Major Mobile Radio Systems**
  - 1934 - Police Radio uses conventional AM mobile communication system.
  - 1935 - Edwin Armstrong demonstrate FM
  - 1946 - First public mobile telephone service – Half duplex, push-to-talk
  - 1960 - Improved Mobile Telephone Service, IMTS - full duplex
  - 1960 - Bell Lab introduce the concept of Cellular mobile system
  - 1968 - AT&T propose the concept of Cellular mobile system to FCC.
  - 1976 - Bell Mobile Phone service, poor service due to call blocking
  - 1983 - Advanced Mobile Phone System (AMPS), FDMA, FM
  - 1991 - Global System for Mobile (GSM), TDMA, GMSK
  - 1991 - U.S. Digital Cellular (USDC) IS-54, TDMA, DQPSK
  - 1993 - IS-95, CDMA, QPSK, BPSK

Example of Mobile Radio Systems

- Cordless phone
- Remote controller (TV remote, Car Remote)
- Hand-held walkie-talkies
- Pagers
- Cellular telephone
- Wireless LAN

Example of Mobile Radio Systems

Pager
- receive only
- tiny displays
- simple text messages

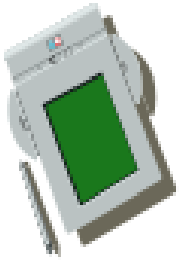
PDA
- simple graphical displays
- character recognition
- simplified WWW

Laptop
- fully functional
- standard applications

Sensors, embedded controllers











Mobile phones
- voice, data
- simple text displays

Palmtop
- tiny keyboard
- simple versions of standard applications

Paging Systems

- Conventional paging system are wireless communication systems used to **send brief messages** to a subscriber.

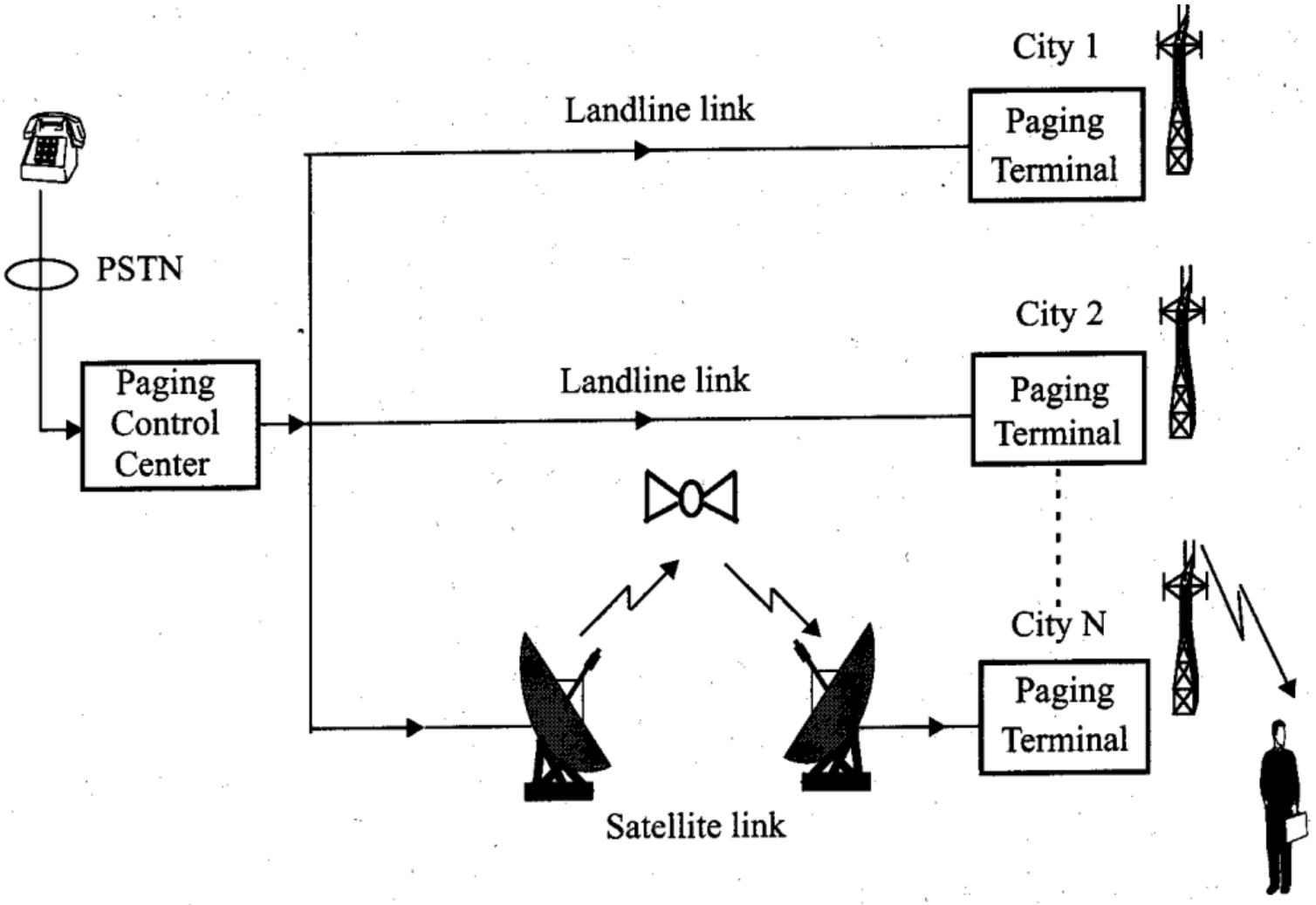
- Depending upon the type of service required, the message may be either **a numeric , an alphanumeric or a voice** message.

- Paging systems are used to notify a subscriber of **the need to call** a particular telephone number or **travel to a particular location** to receive further instructions.

- In Modern paging system: news headline, stock quotations, faxes, etc. may be sent.

Paging system

- Paging receivers are simple and inexpensive, but the transmission system required is quite sophisticated.
- designed to provide ultra-reliable coverage, even **inside buildings**
- Buildings can attenuate radio signals by 20 or 30 dB, making the choice of base station locations difficult for the paging companies.
- Small RF bandwidths are used to maximize the signal-to-noise ratio at each paging receiver, so low data rates (6400 bps or less) are used.


City 1
Landline link
Paging Terminal
PSTN
City 2
Paging Control Center
Landline link
Paging Terminal
City N
Paging Terminal
Satellite link

Types of paging

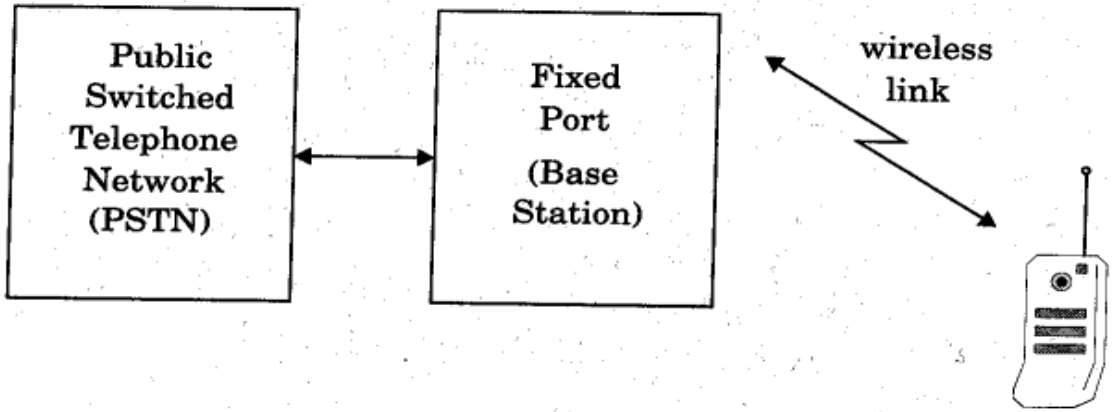
- There are two different kinds of paging Systems: **limited range** and **wide-area**.

- As the name suggests, limited-range paging sends messages over a relatively **small area using a low-powered transmitter**. For example It's perfect for sending emergency messages to all the doctors in a hospital.

- Wide-area paging **is more like national radio broadcasting** . A system of radio transmitters sends pager messages across a whole country in hopes that you'll be somewhere near one of them. In the UK, for example, the wide-area paging network uses something like 500 transmitter antennas—more than enough to cover a country that size.

Cordless Telephone System

- Cordless telephone systems are **full duplex communication systems**
- First generation cordless phone
  - in-home use
  - communication to dedicated base unit
  - few tens of meters
- Second generation cordless phone
  - outdoor
  - combine with paging system
  - few hundred meters per station

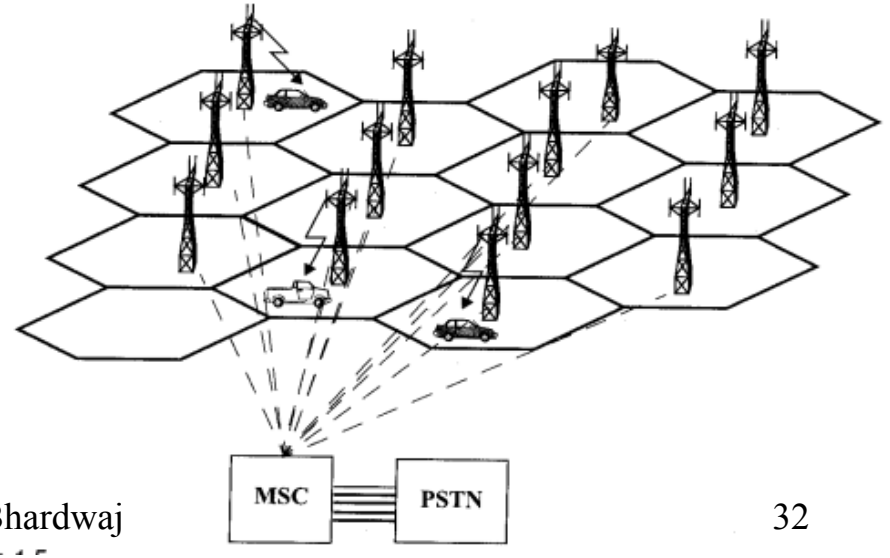


Explanation cordless

- Cordless telephone systems are full duplex communication systems that uses radio to connect a portable handset to a dedicated base station, which is the connected to a dedicated telephone line with a specific telephone number on public switched telephone network (PSTN).
- In first generation cordless telephone systems, the portable unit communicates only to the dedicated base unit and only over distances of few tens of meters.
- Second generation cordless telephones have recently been introduced which allows subscribers to use their handsets at many outdoors locations within urban centers such as London or Hong kong.
- Modern cordless telephones are sometimes connected with paging receivers so that the subscriber may first be paged and then respond to the page using the cordless telephone.
- cordless telephones provides the user with limited range and mobility as it is not possible to maintain the call if user travels outside the range of the base station

Cellular Telephone Systems

- Provide connection to the PSTN for any user location within the radio range of the system.
- Characteristic
  - Large number of users
  - Large Geographic area
  - Limited frequency spectrum
  - Reuse of the radio frequency by the concept of "cell''.
- Basic cellular system: mobile stations, base stations, and mobile switching center.



Evolution of cellular system

- the evolution of cellular networks from the perspective of the multiple access schemes because at the end of the day, we have to support a large number of users. That's where the revenue is going to come from which will ultimately propel the growth. So we start with the first generation.

Cellular networks: Evolution

- **First Generation (1G)**
  - Launched in the mid 1980's.
  - Purely analog.
  - Analog modulation mostly FM
  - Voice traffic only
  - FDMA multiple access schemes
  - Confined to national boundaries (limited users)
  - Example is the AMPS (Advanced mobile phones services)

Explanation – 1G

- It was **launched in the mid 1980's**. They were **purely analog**. They used **analog modulation mostly frequency modulation (FM).** They were intended primarily for **voice traffic only** because at that time phones meant voice. They would use **FDMA multiple access schemes**, there had to be a multiple access. So they would just chop up the entire frequency band into sub bands and use analog transmission in each sub band. They were **confined to national boundaries** only. It's an important thing because if you confine your phone service to a local territory, you'll **limit the number of users and customers**. An example is the **AMPS** or the advanced mobile phones services popular in the US in the mid nineteen eighty1980"s. As cellular networks evolved, the second generation came into being.

Cellular networks: Evolution

- **Second Generation (2G)**
  - Digital – used digital modulation
  - objective- Voice traffic only , but also supported 10kbps data rate
  - TDMA/FDMA/CDMA
  - No internet access
- **E.g.- GSM(Global system for mobile communication)**
  - GMSK and TDMA/FDMA
  - 900MHZ and 1800MHZ band
- **Personal Digital communications(PDC)**
  - Popular in Japan
- **Interim Standard 95 (IS-95)**
  - CDMA
  - US/ South korea
  - **At the end , data was gaining momentum –** need improvement, sold enough 2G

Explanation 2G

- 2G was developed also for **voice communication again**. At that time it was unthinkable that you will be using **phones to communicate data**. Why would anybody want to send data using the phone? That was the third process. Because nobody believed that you would be accessing the **internet through the phone**. Internet itself was in its earlier stages. But what was different is the **2G systems were digital**. Digital in the sense that they used a **digital modulation**. **GSM** for example, it's a 2G phone system. It uses **GMSK**- the Gaussian minimum shift keying it's a digital modulation technique. For multiple access they used **TDMA FDMA and CDMA**. They could provide data rates but of the order of about **10 kbps**. So even though they were designed for voice, there was a provision to handle data simply because they were digital schemes. But their objective was for voice communications. Clearly at the end of 2G systems, people realized that **data traffic is gaining momentum**. Something has to be done to improve the data carrying capability. **But by that time, they had sold enough 2G mobile phones** to completely change the standards immediately. So they had to do something as an add-on. so they sat together and tried to evolve something called as a 2.5 G.

Explanation- 2G examples

- The first and the foremost example is the **GSM** "global system for mobile communications". It uses a combination of **TDMA and FDMA**. It was initially **started at 900 MHz. now you also have an 1800 MHz** variant.
- Another popular 2G system which was popular **only in Japan is the PDC** or the "personal digital communications". It was very popular but the Japanese couldn't sell it elsewhere and therefore Japan is most excited to get into the 3G race. The third generation systems are not going to be confined to any global territory or national boundary. So Japan missed out on getting a big chunk of the 2G market. But it wants to get a head start in the 3G market.
- **IS 95** was different. It was **CDMA** based the code division multiple access and **Qualcomm** the company which essentially propose CDMA was fueling this growth. It was **popular in US and South Korea**.

2G Technologies - Comparasion

	cdmaOne (IS-95)	GSM, DCS-1900	IS-54/IS-136 PDC
Uplink Frequencies (MHz)	824-849 (Cellular) 1850-1910 (US PCS)	890-915 MHz (Eurpe) 1850-1910 (US PCS)	800 MHz, 1500 Mhz (Japan) 1850-1910 (US PCS)
Downlink Frequencies	869-894 MHz (US Cellular) 1930-1990 MHz (US PCS)	935-960 (Europa) 1930-1990 (US PCS)	869-894 MHz (Cellular) 1930-1990 (US PCS) 800 MHz, 1500 MHz (Japan)
Deplexing	FDD	FDD	FDD
Multiple Access	CDMA	TDMA	TDMA
Modulation	BPSK with Quadrature Spreading	GMSK with BT=0.3	$\pi/4$ DQPSK
Carrier Seperation	1.25 MHz	200 KHz	30 KHz (IS-136) (25 KHz PDC)
Channel Data Rate	1.2288 Mchips/sec	270.833 Kbps	48.6 Kbps (IS-136) 42 Kbps (PDC)
Voice Channels per carrier	64	8	3
Speech Coding	CELP at 13Kbps EVRC at 8Kbps	RPE-LTP at 13 Kbps	VSELP at 7.95 Kbps

Cellular networks: Evolution

- **Limitations of Second Generation (2G)**
  - Developed for voice communications (unsuitable for data traffic)
  - Average data rates were the order of only tens of Kbps (kilobits per second)
  - Not suitable for internet (not a packet switched service)
  - Multiple standards (no true global coverage)

Explanation

- So what are the major impediments that we see today? First of all it was developed for **voice communication** and implicit statement **unsuitable for data traffic**, although data traffic could be carried out. The average data rates were the order of **only tens of kilobits per second**. It's not good to the customers who have become used to the luxury of high data rates. It is definitely **not suitable for internet** because the original GSM phone was a circuit switched system and **not a packet switched service**. But for internet we have to have some kind of a packet switched network. Then a general problem with all second generation systems was they were **too many standards**. Although GSM took the major chunk of the pie, still they were too many standards one in the US European standard, Japanese standard etc. so it was a mess. All these motivated toward the 2.5 G. it resulted as an effort to remove the impediments of the 2G systems.

Cellular networks: Evolution

- **2.5 G**
  - The effort to remove the impediments of the 2G systems resulted in 2.5 G
  - Digital system
  - Voice+ low data rate traffic
  - Internet access can be done through GPRS or general packet radio service
  - EDGE (enhanced data rates for global evolution) uses better digital modulation techniques

Explanation 2.5 G

- It is again a **digital system**. It is designed this **time not only for voice but for low data rate traffic**. So we make it explicit that it's going to be also for data traffic. **Internet access can be done through** GPRS or general packet radio service . So it is the first step of going towards 3G and then the step two called the **EDGE (enhanced data rates for global evolution)** which **uses better digital modulation techniques** to get that EDGE. So if you look at the steps you have a 2 G, then you can go to GPRS and then proceed on to EDGE and finally to the 3 G systems. That is the road map that has been laid out. Most of the GSM phones today in India are 2.5 G phones. The initial CDMA phones that have come out CDMA 2001 X EV DO are the tip of the ice berg for the 3G phones. They will write it as 3G enabled but truly all the parameters required for 3G are not being satisfied.

Cellular networks: Evolution

- **3G**
  - Digital modulation
  - Simultaneous voice + high speed data.
  - Multi-mega bit internet, thrust internet
  - voice activated calls
  - Multimedia Transmission
  - World wide standard
  - E.g. W-CDMA and CDMA 2000

Explanation 3G

- So look at the evolution of cellular networks. The 3G again requires you to use **digital modulation**. but **apart from voice, we have high speed data**. You have **multi-mega bit internet** access. So there's a **thrust internet**. I can check my emails while driving in my car at 80 km/hr. then the **voice activated calls** and your multimedia transmission are there. So I can download a movie clip and watch on my phone while walking around. So 3G puts constraints on how fast you go and how fast you can download the traffic. Clearly if you move much faster, the channel becomes more difficult to handle. The associated problem is fast fading Doppler effects which have to be taken care of. So whenever you talk about 3G and the quality of service you have to have an access, in which you tell how mobile you are. It's not just a mobile phone but how fast you are moving. Are you stationary? Are you a pedestrian traffic or are you in a vehicle? Is it a vehicular traffic? For all these things, you need to have a certain kind of quality of service. So 3G is still coming. Trials are being run. Initial three 3G phones have been released. But people have started talking about something more which is the 4G. One important thing about 3G is it is truly a world standard that is evolving. There are two standards which have been debated. One is the W-CDMA which is the wide band CDMA and the other is a CDMA 2000 standard. Both have a lot of similarity and basic differences. I guess both will be incorporated because a lot of money has been poured into the development of each of the W-CDMA and CDMA 2000 systems.

Cellular networks: Evolution

Need of 4G

- Present communication systems are primarily **designed for one specific application** such as mobile phone for voice or **high rate data** for wireless local area networks (WLAN)
- 4G will **integrate various networks, functions and applications**
- The 4 G will **truly create the global information multimedia village**. The “anywhere” “any time” communication communication for all the applications should become possible.
- It will support a variety of data rates at variety of speeds from 2G to 3G to 3G+ WLAN systems to PICO and small microcellular connectivity and fixed line systems.

Explanation 4G

Let's look at the need for 4G that is still several years down the road. **So present communication systems are primarily designed for one specific application**. for the first time 4G systems are looking from the applications point of view. So either it is a **mobile phone for voice or high rate data for local area networks**, 4G systems would like to combine all of this. 4G will integrate various networks, functions and applications. Today if I walk into a room which has a wireless local area network enabled through 802.11 B, I cannot interface my mobile phone with that. if I have a MMS clip or an SMS or a V card which I would like to print out on a printer which is connect to an 802.11 B wireless LAN, I have a problem. 4 G will solve that problem. I can walk into the room, configure my phone to send the printout to a printer connected to some other network, may be 802.11 B or an 802.15.3 UWB network and then I should be able to get my printout. **The 4 G will truly create the global information multimedia village**. The "anywhere" "any time" communication for all the applications should become possible. It should support a variety of data rates at variety of speeds. There will be various sizes of cells. Remember the cellular concept came from originally the mobile phone users where we had the city divided into cells. Each cell with a base station. Today we are talking about not just the macro cells but micro cells, Pico cells, home cells and even body area networks. So the other interesting feature about 4 G is that we will look at various sizes of cells. Each cell for a different application. Each application with a different quality of service and each application with a different data rate. 4G promises to combine all of these things but still it is in the future

Cellular Networks- Summarized evolution

- First Generation
  - Analog Systems
  - Analog Modulation, mostly FM
  - AMPS
  - Voice Traffic
  - FDMA/FDD multiple access
- Second Generation (2G)
  - Digital Systems
  - Digital Modulation
  - Voice Traffic
  - TDMA/FDD and CDMA/FDD multiple access
- 2.5G
  - Digital Systems
  - Voice + Low-datarate Data
- Third Generation
  - Digital
  - Voice + High-datarate Data
  - Multimedia Transmission also

Table Comparison of Mobile Communication Systems—Mobile Station

Service	Coverage Range	Required Infra-structure	Complexity	Hardware Cost	Carrier Frequency	Functionality
TV Remote Control	Low	Low	Low	Low	Infrared	Transmitter
Garage Door Opener	Low	Low	Low	Low	< 100 MHz	Transmitter
Paging System	High	High	Low	Low	< 1 GHz	Receiver
Cordless Phone	Low	Low	Moderate	Low	< 1 GHz	Transceiver
Cellular Phone	High	High	High	Moderate	< 2 GHz	Transceiver

Table Comparison of Mobile Communication Systems—Base Station

Service	Coverage Range	Required Infra-structure	Complexity	Hardware Cost	Carrier Frequency	Functionality
TV Remote Control	Low	Low	Low	Low	Infrared	Receiver
Garage Door Opener	Low	Low	Low	Low	< 100 MHz	Receiver
Paging System	High	High	High	High	< 1 GHz	Transmitter
Cordless Phone	Low	Low	Low	Moderate	< 1 GHz	Transceiver
Cellular Phone	High	High	High	High	< 2 GHz	Transceiver